

When does label smoothing help? Ly muller demonstrates that while label smoothing improves model generalization and calibration by reducing over-confidence. It hampers knowledge distillation by destroying inter-class similarities. The study finds that smoothed labels create tight, isolated representation clusters making the model a poor teacher for subsequent distillation tasks.

Accuracy:

I have done experiment on label smoothing on base task. The model with normal training have efficiency of 84.87%. The model with label smoothing achieve final efficiency of 85.05%.

Loss:

The model trained with label smoothing have high value at initial stage but remains relatively stable and converges to a similar very low value.

Different Smoothing value:

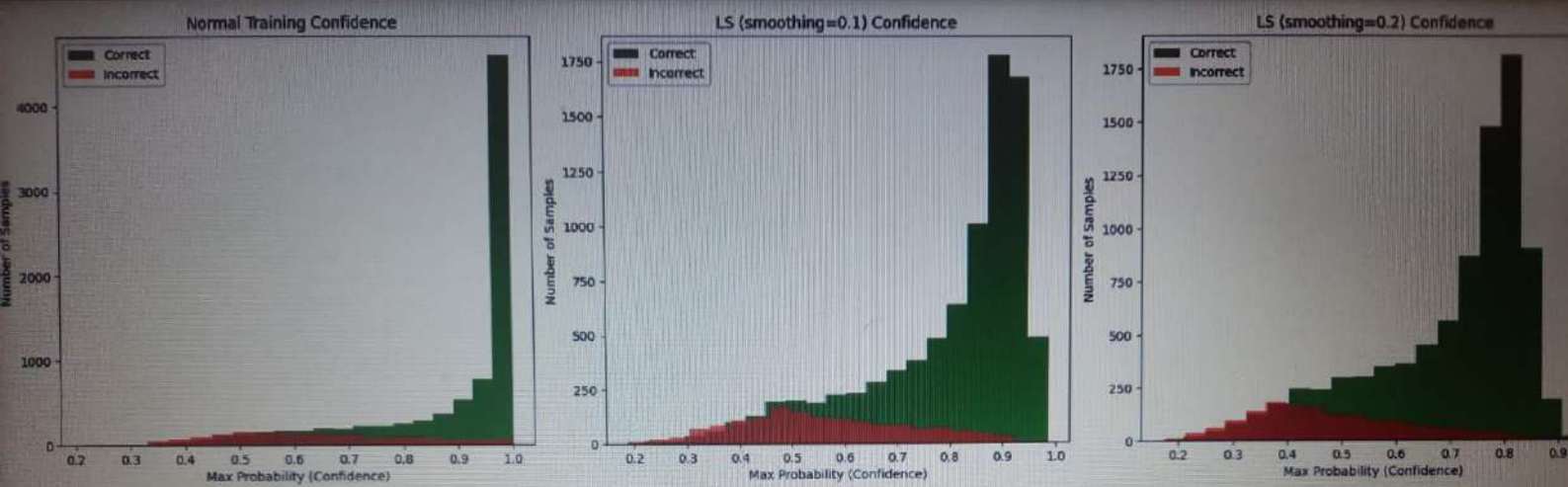
The model is trained with normal smoothing at 0.1 and 0.2

Accuracy at smoothing = 0.1 is 84.95%.

Accuracy at smoothing = 0.2 is 85.01%.

Both the smoothing variant (0.1 and 0.2) have higher losses than normal training because label smoothing encourage

the model to be less confident in his prediction



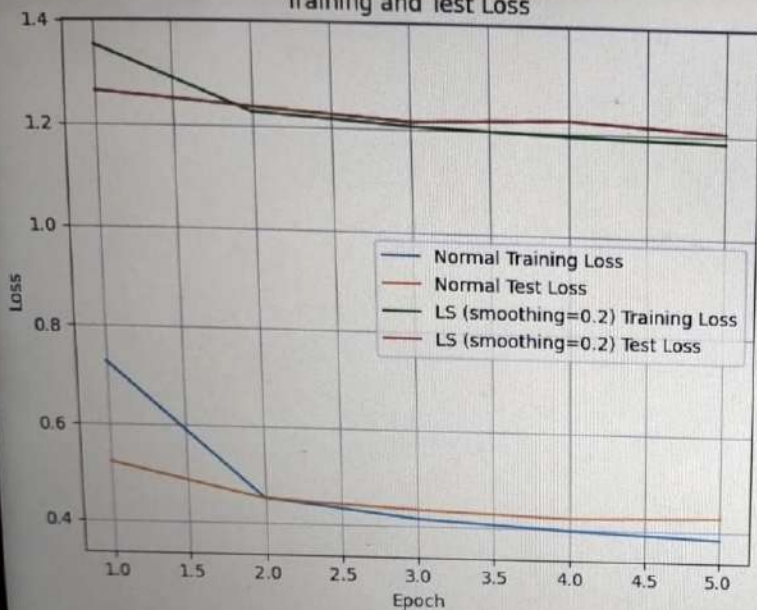
Average Confidence for Correct Predictions:

Normal: 0.8940
LS (smoothing=0.1): 0.7995
LS (smoothing=0.2): 0.7027

Average Confidence for Incorrect Predictions:

Normal: 0.6310
LS (smoothing=0.1): 0.5581
LS (smoothing=0.2): 0.4775

Training and Test Loss



Final Normal Test Accuracy: 84.87%
Final Label Smoothing (smoothing=0.2) Test Accuracy: 85.41%

Training and Test Accuracy

